

AI 504 activity: Canonical models

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Week 3 recitation

1. Work over the set of nouns $\mathbf{P} = \{p, q, r\}$. Let $\Gamma = \{\text{All } p \text{ are } q, \text{All } p \text{ are } r\}$. Compute each of the following three sets:
 - $\{x \in \mathbf{P} : \Gamma \vdash \text{All } x \text{ are } p\} =$
 - $\{x \in \mathbf{P} : \Gamma \vdash \text{All } x \text{ are } q\} =$
 - $\{x \in \mathbf{P} : \Gamma \vdash \text{All } x \text{ are } r\} =$
2. Now construct a $\mathbf{P}$ -model whose *domain is also* $\mathbf{P}$, and where the interpretation of p, q, r are the three sets above respectively. (Caution: $\mathbf{P}$ plays two roles here: it is both the domain as well as the underlying signature. While this may be momentarily confusing, there is nothing wrong with it!) Draw your answer below.
3. Look at your model. What sentences does it satisfy? Write them all below. Does that set contain Γ ?
4. Now go exactly through the same process, but for the set $\Sigma = \{\text{All } p \text{ are } q, \text{All } r \text{ are } p\}$. Draw the resulting model below.
5. Again, write the set of sentences that your model satisfies. Does it contain Σ as a subset?

- [illegible]

9. Again, let $\mathbf{P}$ be any signature and Γ be any set of $\mathbf{P}$ -sentences. Using your definition of the canonical model $\mathcal{M}_\Gamma$ from the previous problem, rigorously prove that $\mathcal{M}_\Gamma \models \Gamma$; i.e., that $\mathcal{M}_\Gamma$ satisfies every sentence in Γ .

10. Let $\mathbf{P}$ be any signature, Γ be any set of $\mathbf{P}$ -sentences, *and* let φ be any arbitrary $\mathbf{P}$ -sentence. Suppose that $\mathcal{M}_\Gamma \models \varphi$. Prove that $\Gamma \vdash \varphi$.